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Topic : A Neural Network–Based Structural Gravity Model of EU–USA Free Trade Agreement Spillovers on India

Dataset Justification & Documentation

1. Dataset: CEPII Gravity Database

- **Description:** A comprehensive dataset designed specifically for estimating structural gravity models. It provides consistent data on bilateral trade, along with extensive geographic, cultural, and historical linkage metrics.
- **Source:** Centre d'Études Prospectives et d'Informations Internationales (CEPII).
- **Time Period:** 1948–2020.
- **Key Variables:** Contiguity, common official language, colonial linkages, population-weighted distance, and historical diplomatic ties.
- **Justification:** This is the foundational dataset for the spatial component of your model. To accurately compute the multilateral resistance terms using your Graph Neural Network, the GNN requires these exact cultural and geographic friction variables as edge weights between country nodes.
- **URL:** https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele.asp

2. Dataset: International Trade and Production Database for Estimation (ITPD-E) Release 3

- **Description:** A uniquely structured database containing both international and *domestic* trade data across agriculture, mining, manufacturing, and services.
- **Source:** United States International Trade Commission (USITC).
- **Time Period:** 1986–2022.
- **Key Variables:** Bilateral trade flows, domestic sales, and industry-level production spanning 170 industries across 264 countries.
- **Justification:** Modern structural gravity estimations mathematically require intra-national trade data (a country trading with itself) to solve the "distance puzzle" and establish a baseline for border effects. Your XGBoost and LSTM models will overfit on standard international-only data; the ITPD-E prevents this by providing the necessary domestic baselines.
- **URL:** https://gravity.usitc.gov/

3. Dataset: WTO Tariff & Trade Data (Integrated Data Base - IDB)

- **Description:** The official repository for applied tariffs, bound duties, and import data notified by WTO Members, integrating data on Regional Trade Agreements (RTAs).
- **Source:** World Trade Organization (WTO).

- **Time Period:** 1996–Present.
- **Key Variables:** MFN applied tariffs, bilateral preferential tariffs, and Free Trade Agreement (FTA) status indicators at the HS6 product level.
- **Justification:** To answer your second Research Question (RQ2) regarding spillover effects, your Causal Forest model requires a precise, exogenous policy shock variable. This dataset provides the exact bilateral tariff variations between the EU, USA, and India needed to isolate the causal treatment effect of the FTA.
- **URL:** https://ttd.wto.org/en

4. Dataset: World Development Indicators (WDI)

- **Description:** The primary World Bank collection of development indicators, compiled from officially recognized international sources.
- **Source:** The World Bank.
- **Time Period:** 1960–Present.
- **Key Variables:** GDP (current US$), GDP per capita, inflation rates, and real effective exchange rates.
- **Justification:** These variables represent the "economic mass" (supply capability of the exporter and demand capacity of the importer). In your architecture, these will serve as the core node-level features that feed into both the LSTM for time-series forecasting and the Reinforcement Learning agent for state-space representation.
- **URL:** https://databank.worldbank.org/source/world-development-indicators

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List of 10 Relevant Research Papers (For your PDF Zip Folder)

To ensure your literature review reflects state-of-the-art AI applications in international trade, locate and download the PDFs for these specific papers. They directly align with your methodology (GNNs, Causal ML, XGBoost, and Gravity Models).

1. **Verstyuk, S. (2022).** *Machine Learning the Gravity Equation for International Trade.* (Directly addresses RQ1 and RQ3; formulates Graphical Neural Networks for trade).
2. **Gopinath, M., & Ba, F. A. (2020).** *Machine Learning in Gravity Models: An Application to Agricultural Trade.* NBER Working Paper. (Compares supervised/unsupervised ML with classical gravity).
3. **Fernandes, A. M., et al. (2021).** *Machine Learning in International Trade Research: Evaluating the Impact of Trade Agreements.* World Bank Group. (Crucial for using AI to isolate FTA policy effects).
4. **USITC Working Paper (2023).** *Neural Network Analysis of International Trade.* (Benchmarks neural networks against PPML and OLS baselines).
5. **Olofsson, et al. (2025).** *An evaluation of gravity models and artificial neuronal networks on bilateral trade flows.* (Recent benchmark comparing Feedforward Neural Networks to deterministic gravity).

6. **Yotov, Y. V., et al. (2016).** *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model.* WTO/UNCTAD. (The definitive theoretical baseline against which you must compare your AI models).
7. **Athey, S., & Wager, S. (2019).** *Estimating Treatment Effects with Causal Forests: An Application.* (Methodological foundation for your Causal Forest implementation to measure EU-USA spillovers).
8. **Lundberg, S. M., & Lee, S. I. (2017).** *A Unified Approach to Interpreting Model Predictions.* (The foundational paper for SHAP; required to justify your explainable AI approach in RQ3).
9. **Vannoorenberghe, G., et al. (2022).** *Spillover Effects of Free Trade Agreements in Global Value Chains.* (Provides the economic theory framework for third-party effects on nations like India).
10. **Wu, Z., et al. (2020).** *A Comprehensive Survey on Graph Neural Networks.* (Essential methodological backing for using GNNs to map the topological multilateral resistance in global trade).

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1. Common Variables for Dataset Integration (Merging Keys)

To construct your multi-dimensional panel dataset, you will perform relational merges (e.g., SQL JOIN or Pandas merge) across the CEPII, ITPD-E, WTO, and WDI datasets. The following key variables are common across these platforms and will serve as your primary merging keys:

- **Exporter Country Code (iso3_o / exporter):** The standardized 3-letter ISO 3166-1 alpha-3 code representing the origin country (e.g., "IND" for India, "USA" for the United States). WDI, CEPII, and ITPD-E universally support ISO-3.
- **Importer Country Code (iso3_d / importer):** The 3-letter ISO 3166-1 alpha-3 code representing the destination country.
  - *Note on Merging:* WDI provides unilateral data (one country per row), so it must be merged twice—once on the iso3_o to attach exporter characteristics (like Exporter GDP) and once on the iso3_d to attach importer characteristics (like Importer GDP).
- **Time Period (year):** The 4-digit year format (e.g., 2018). Since trade dynamics and policies evolve, the temporal dimension is crucial for the LSTM component of your model.
- **Sector/Product Code (e.g., HS6 or ISIC):** While CEPII and WDI provide aggregate macroeconomic data, the WTO Tariff data and ITPD-E offer granular sector-level data. You will use harmonization tables to align HS (Harmonized System) product codes with broad economic sectors to merge tariff shocks accurately with corresponding trade flows.

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2. How the Datasets Will Be Used in the Study

Each dataset feeds directly into a specific architectural component of your hybrid AI model. Here is the operational plan for your data:

A. International Trade and Production Database (ITPD-E)

- **Usage:** This acts as your **Ground Truth / Target Variable**. It provides the actual bilateral trade flows and domestic trade flows.
- **AI Integration:** The historical trade flows from this dataset will be formatted as time-series sequences and fed into the **LSTM** to learn temporal persistence. The XGBoost model will use this as the target variable to minimize prediction error against.

B. CEPII Gravity Database

- **Usage:** This provides the structural frictions (distance, contiguity, language) required by gravity theory.
- **AI Integration:** These variables will define the **Edge Weights and Adjacency Matrix** for your **Graph Neural Network (GNN)**. Instead of just plugging "distance" into a regression, the GNN will use these geographical/historical ties to map the complex network of "multilateral resistance"—how a shock between the EU and USA cascades through the network to affect India.

C. World Development Indicators (WDI)

- **Usage:** This provides the macroeconomic "mass" variables (GDP, inflation, exchange rates) that dictate supply and demand capacities.
- **AI Integration:** These variables serve two purposes. First, they act as the **Node Features** for the countries in the GNN. Second, they define the **State Environment** for the **Reinforcement Learning (RL)** agent, allowing it to understand the macroeconomic context before suggesting a policy action.

D. WTO Tariff & Trade Data (IDB)

- **Usage:** This provides the exogenous policy shock metrics (applied tariffs, FTA status).
- **AI Integration:** This is the core engine for your **Causal Forests** and **SHAP** analysis. The WTO data provides the "Treatment Variable" (the EU-USA FTA tariff drop). The Causal Forest will use this treatment to estimate the heterogeneous spillover effects on India (answering RQ2). Later, SHAP will isolate the specific explanatory weight of these tariff variables versus standard distance variables to interpret the model's logic.

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1. Problem Statement

The prospective Free Trade Agreement (FTA) between the European Union (EU) and the United States (USA) threatens to significantly alter global value chains. For third-party emerging economies like India, this macroeconomic shock presents a critical vulnerability: the potential for severe **trade diversion**, wherein EU and US importers substitute Indian goods with tariff-free products from their new FTA partner (Vannoorenberghe et al., 2022).

While the Structural Gravity Model is the workhorse of international trade policy analysis (Yotov et al., 2016), classical econometric estimation techniques—such as Ordinary Least Squares (OLS) and Poisson Pseudo Maximum Likelihood (PPML)—suffer from critical limitations. They enforce rigid, linear functional forms, struggle to map complex spatial network dependencies (multilateral resistance), and often fail to accurately isolate the heterogeneous, third-party spillover effects of FTAs. Consequently, policymakers lack precise, dynamic, and non-linear forecasting tools to quantify India's specific sectoral vulnerabilities and formulate optimal, data-driven countermeasures. This study addresses this gap by replacing classical econometric estimations with a highly non-linear, causally aware Artificial Intelligence architecture.

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2. Methodology and Model Explanation

This research employs a novel, hybrid AI-Econometric framework that maps the theoretical constraints of structural gravity into a multi-layered machine learning pipeline. The theoretical foundation is the structural gravity equation (Anderson & van Wincoop, 2003):

Where trade flows depend on economic mass , bilateral trade costs, and unobservable multilateral resistance terms . The methodology operationalizes this equation through the following AI components:

A. Spatiotemporal Network Modeling (GNN + LSTM)

Standard gravity models use dummy variables (fixed effects) to control for multilateral resistance, which computationally limits dynamic network analysis. This study utilizes **Graph Neural Networks (GNNs)** to model the global trade network natively as a mathematical graph (Wu et al., 2020). Countries are nodes containing macroeconomic features (GDP, inflation), and bilateral frictions (distance, common language) serve as edge weights. The GNN maps the topological trade dependencies, while a **Long Short-Term Memory (LSTM)** network processes the sequential historical data to capture the time-dependent persistence of supply chains.

B. Non-Linear Gravity Estimation (XGBoost)

To estimate the complex, non-linear interactions between gravity variables that OLS and PPML miss, the study employs **XGBoost** (Extreme Gradient Boosting). Recent literature demonstrates that tree-based gradient boosting significantly outperforms traditional parametric models in minimizing prediction error for bilateral trade flows, particularly when dealing with high-dimensional trade friction data (Gopinath & Ba, 2020).

C. Causal Inference for Spillover Effects (Causal Forests)

Predictive machine learning cannot inherently answer "what if" policy questions. To isolate the specific spillover effect of the EU-USA FTA on India (RQ2), the study utilizes **Causal Forests**, an advanced causal machine learning technique (Athey & Wager, 2019). By treating the FTA implementation as a policy shock (treatment), the Causal Forest will estimate the Conditional Average Treatment Effect (CATE), identifying the precise heterogeneous impact on Indian export sectors while controlling for confounding variables.

D. Policy Optimization (Reinforcement Learning)

To move beyond impact assessment into actionable policy prescription, a **Reinforcement Learning (RL)** agent will be deployed. The environment will be defined by the post-FTA trade network states. The RL agent, acting as the Indian policymaker, will explore adjusting domestic tariffs and export subsidies (actions) to maximize India's net export volume and minimize trade diversion (rewards).

E. Explainable AI (SHAP)

A primary criticism of neural networks in economics is their "black box" nature. To ensure the model is robust and the findings are interpretable for policymakers (RQ3), **SHAP (SHapley Additive exPlanations)** values will be integrated (Lundberg & Lee, 2017). SHAP will deconstruct the AI's predictions, calculating the exact marginal contribution of each variable (e.g., quantifying exactly how much of a predicted drop in Indian exports is driven by the FTA tariff change versus geographical distance).

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3. Literature Cited in Methodology

- **Anderson, J. E., & van Wincoop, E. (2003).** Gravity with Gravitas: A Solution to the Border Puzzle. *American Economic Review*, 93(1), 170-192. *(Provides the theoretical structural gravity equation).*
- **Athey, S., & Wager, S. (2019).** Estimating Treatment Effects with Causal Forests: An Application. *Observational Studies*, 5(2), 36-51. *(Justifies the use of Causal Forests for FTA policy shocks).*
- **Gopinath, M., & Ba, F. A. (2020).** Machine Learning in Gravity Models: An Application to Agricultural Trade. *NBER Working Paper No. 27151*. *(Supports the use of XGBoost over traditional models).*
- **Lundberg, S. M., & Lee, S. I. (2017).** A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, 30. *(Foundational backing for using SHAP in policy analysis).*

- **Vannoorenberghe, G., et al. (2022).** Spillover Effects of Free Trade Agreements in Global Value Chains. *World Bank Group. (Highlights the specific problem of trade diversion for third-party countries).*
- **Wu, Z., et al. (2020).** A Comprehensive Survey on Graph Neural Networks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(1), 4-24. *(Methodological support for mapping complex network structures).*
- **Yotov, Y. V., et al. (2016).** *An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model.* WTO/UNCTAD. *(Establishes the classical PPML/OLS baseline that the AI models will be compared against).*

Literature Review of all the 10 papers related to the topic and form a base for the project :

Title	Author(s)	Year	Problem Statement	Methodology	Results	Future Work
A Comprehensive Survey on Graph Neural Networks	Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Yu, P. S.	2020	Standard deep learning models struggle to process complex, non-Euclidean network topologies (like global trade networks).	Categorizes GNNs into a comprehensive taxonomy (recurrent, convolutional, autoencoders, and spatial-temporal GNNs).	GNNs achieve state-of-the-art accuracy in node classification and network link predictions.	Focus on increasing model depth, scaling for massive dynamic graphs, and improving model interpretability.

Estimating Treatment Effects with Causal Forests: An Application	Athey, S., & Wager, S.	2019	A need for robust algorithms to estimate heterogeneous, non-linear treatment effects (causal impacts) in observational datasets with confounding variables.	Causal Forests (an adaptation of random forests) utilizing orthogonalization and "honest" sample splitting.	Causal forests successfully control for confounding via propensity scores and efficiently identify subgroups experiencing varying causal impacts.	Enhancing the algorithm's capability to handle high-dimensional clustered errors and complex panel data.
A Survey on Graph Neural Networks for Remaining Useful Life Prediction	Wang, Y., Wu, M., Li, X., Xie, L., & Chen, Z.	2024	Predicting longevity and failure in complex systems is challenging due to intricate spatial and temporal sensor	Benchmarking of temporal and spatial-temporal GNNs across four structural adaptation stages.	Spatio-temporal GNNs consistently outperform traditional machine learning and standard deep learning algorithms in dynamic prediction accuracy.	Development of physics-informed GNNs, dynamic graph topology learning, and unsupervised network learning.

			dependencies.			
Deep Integration in Trade Agreements: Labor Clauses, Tariffs, and Trade Flows	Robertson, R.	2021	The impact of "deep" non-tariff clauses (like labor standards) in trade agreements on bilateral trade volume is highly debated and potentially negative.	Pseudo-Poisson Maximum Likelihood (PPML) estimation with a full set of fixed effects to control for endogeneity.	While trade agreements broadly increase trade, the marginal relationship of specific labor clauses is generally negative, causing some trade diversion.	Exploring the granular legal enforcement mechanisms of deep clauses and their specific sectoral impacts.
Determinants of PTA Design: Insights from Machine Learning	Gordeev, S., & Steinbach, S.	2024	Determining exactly which economic, political, and geographic factors drive the inclusion of specific provision	Supervised machine learning (Random Forests) used to handle the high dimensionality of 287 country-pair attributes.	Market contagion, geographic proximity, and domestic governance quality are identified as the most critical determinants	Causal exploration of the individual socio-economic mechanisms highlighted by the ML variable importance metrics.

			s in Preferential Trade Agreements (PTAs).		of PTA design.	
An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model	Yotov, Y. V., Piermartini, R., Monteiro, J.-A., & Larch, M.	2016	Policymakers require a rigorous, practical, and theoretically sound baseline for analyzing and quantifying the effects of trade policies.	Structural gravity modeling estimated via PPML, utilizing intra-national trade data and directional fixed effects.	Provides definitive solutions to the "distance puzzle" and sets the econometric baseline for estimating Regional Trade Agreement (RTA) effects.	Extending structural gravity frameworks to dynamic general equilibrium models to trace long-term economic adjustments.

Machine Learning the Gravity Equation for International Trade	Verstyuk, S., & Douglas, M. R.	2022	Applying flexible, non-linear machine learning architectures to trade without violating the theoretical properties of the gravity model.	Graph Neural Networks (GNNs) for structural trade data mapping, combined with symbolic regression for theoretical extraction.	GNNs fit the structural data exceptionally well, and symbolic regression discovers expressions mathematically mimicking traditional market access functions *ab initio*.	Incorporating dynamic edge modeling for panel data over extended time horizons to capture economic shocks.
Machine Learning in International Trade Research: Evaluating the Impact of Trade Agreements	Breinlich, H., Corradi, V., Rocha, N., Ruta, M., Santos Silva, J., & Zylkin, T.	2021	Estimating the effects of hundreds of specific PTA provisions on trade suffers from severe multicollinearity and overfitting in	Data-driven machine learning variable selection methods (Lasso, Ridge, Random Forests) paired with PPML.	Machine learning successfully isolated provisions related to technical barriers, antidumping, and competition policy as the main trade-enhancing factors.	Refining machine learning selection tools to deal with unobserved heterogeneity and strict causal inference in high-dimensional environments.

			classical models.			
A Unified Approach to Interpreting Model Predictions	Lundberg, S. M., & Lee, S. I.	2017	Complex machine learning models (like deep neural networks) achieve high accuracy but act as uninterpretable "black boxes" for decision-makers.	SHAP (SHapley Additive exPlanations), leveraging cooperative game theory to assign a marginal importance value to each feature.	SHAP unifies six previous feature attribution methods and demonstrates significantly better alignment with human intuition regarding feature importance.	Developing faster, model-specific approximation algorithms for SHAP value computation in massive, high-dimensional datasets.
Machine Learning in Gravity Models: An Application to Agricultural Trade	Gopinath, M., Batarseh, F. A., & Beckman, J.	2020	Traditional parametric gravity models struggle with the complex, non-linear interactions determining agricultur	Comparative analysis of traditional econometric models against supervised ML algorithms (Random Forests, Extreme Gradient Boosting).	Machine learning models (especially Gradient Boosting) significantly outperform traditional PPML and OLS models in out-of-sampl	Integration of Deep Learning and Recurrent Neural Networks (RNNs) to further improve non-linear longitudinal predictions.

			al trade flows.		e prediction accuracy.	

1. Alignment of Previous Studies with my Research Topic

My research sits at the intersection of international economics and advanced machine learning. The existing literature validates this approach and directly supports your methodological choices across three primary themes:

Theme A: The Shift from Classical Econometrics to Non-Linear AI in Trade Historically, the Poisson Pseudo Maximum Likelihood (PPML) structural gravity model has been the gold standard for trade policy analysis (Yotov et al., 2016). However, recent studies highlight its limitations in handling complex, high-dimensional interactions. Gopinath et al. (2020) demonstrated that machine learning algorithms (like Gradient Boosting) significantly outperform PPML in predicting trade flows. Furthermore, Verstyuk & Douglas (2022) proved that Graph Neural Networks (GNNs) map the structural dependencies of global trade more effectively than traditional fixed-effect dummy variables.

- **Alignment:** This literature directly justifies your core methodological transition from classical baselines to a hybrid AI architecture (GNNs and XGBoost) to answer whether AI outperforms classical models.

Theme B: Causal Inference and FTA Spillovers Classical models suffer from severe multicollinearity and endogeneity when trying to isolate the specific impacts of trade agreements (Breinlich et al., 2021). To measure the "spillover" effect of an EU-USA FTA on India, predictive AI is insufficient; you need causal AI. Athey & Wager (2019) established Causal Forests as a robust method for estimating heterogeneous treatment effects.

- **Alignment:** By treating the EU-USA FTA as a "treatment" shock, incorporating Causal Forests aligns your study with the vanguard of causal machine learning, allowing you to isolate the precise trade diversion effects on India without the confounding biases of older methods.

Theme C: Overcoming the "Black Box" via Explainable AI (XAI) A persistent critique of neural networks in economics is their lack of interpretability, which makes them difficult for policymakers to trust. Lundberg & Lee (2017) introduced SHAP (SHapley Additive exPlanations) to deconstruct deep learning predictions.

- **Alignment:** Integrating predictive, explainable SHAP AI into your methodology bridges the gap between complex neural network topologies and actionable economic theory. It ensures your model's robustness and directly aligns with the literature's demand for transparent policy tools.

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2. Refined Research Questions (RQs)

To ensure perfect alignment with the nuanced methodologies in the literature (specifically GNNs, Causal Forests, and SHAP), your original RQs should be slightly refined to be more specific and rigorous:

- **Refined RQ1:** To what extent do spatio-temporal Graph Neural Networks (GNNs) mitigate the limitations of classical PPML estimations in mapping the structural gravity of bilateral trade? *(Refines old RQ1/RQ3 by explicitly naming the algorithms being compared based on Verstyuk, 2022).*
- **Refined RQ2:** Utilizing Causal Forests, what are the heterogeneous, causal spillover effects (trade diversion versus trade creation) of an EU-USA Free Trade Agreement on India's export flows? *(Refines old RQ2 by integrating the causal methodology from Athey & Wager, 2019).*
- **Refined RQ3:** How can SHAP-based explainable AI be leveraged to interpret the non-linear macroeconomic drivers of trade flows and validate AI-driven policy optimizations? *(Replaces the redundant original RQ3 with a focus on model interpretability, matching Lundberg & Lee, 2017).*

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3. Refined Research Objectives

Reflecting the updated RQs, your objectives should explicitly state the advanced techniques you are using to address the literature's gaps:

1. **To develop and benchmark** a spatio-temporal Graph Neural Network (GNN) structural gravity model against traditional econometric baselines (OLS, PPML) for predicting global trade flows.
2. **To isolate and quantify** the exogenous spillover effects of an EU-USA FTA on India using Causal Forests to identify specific sectoral vulnerabilities.
3. **To implement** predictive, explainable SHAP AI to extract transparent, theoretical economic insights from the neural network and simulate optimal policy responses using Reinforcement Learning.

References in this Section

- **Athey, S., & Wager, S. (2019).** Estimating Treatment Effects with Causal Forests: An Application. *Observational Studies*.
- **Breinlich, H., et al. (2021).** Machine Learning in International Trade Research: Evaluating the Impact of Trade Agreements. *World Bank Group*.
- **Gopinath, M., et al. (2020).** Machine Learning in Gravity Models: An Application to Agricultural Trade. *NBER*.
- **Lundberg, S. M., & Lee, S. I. (2017).** A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*.
- **Verstyuk, S., & Douglas, M. R. (2022).** Machine Learning the Gravity Equation for International Trade.
- **Yotov, Y. V., et al. (2016).** An Advanced Guide to Trade Policy Analysis: The Structural Gravity Model. *WTO/UNCTAD*.